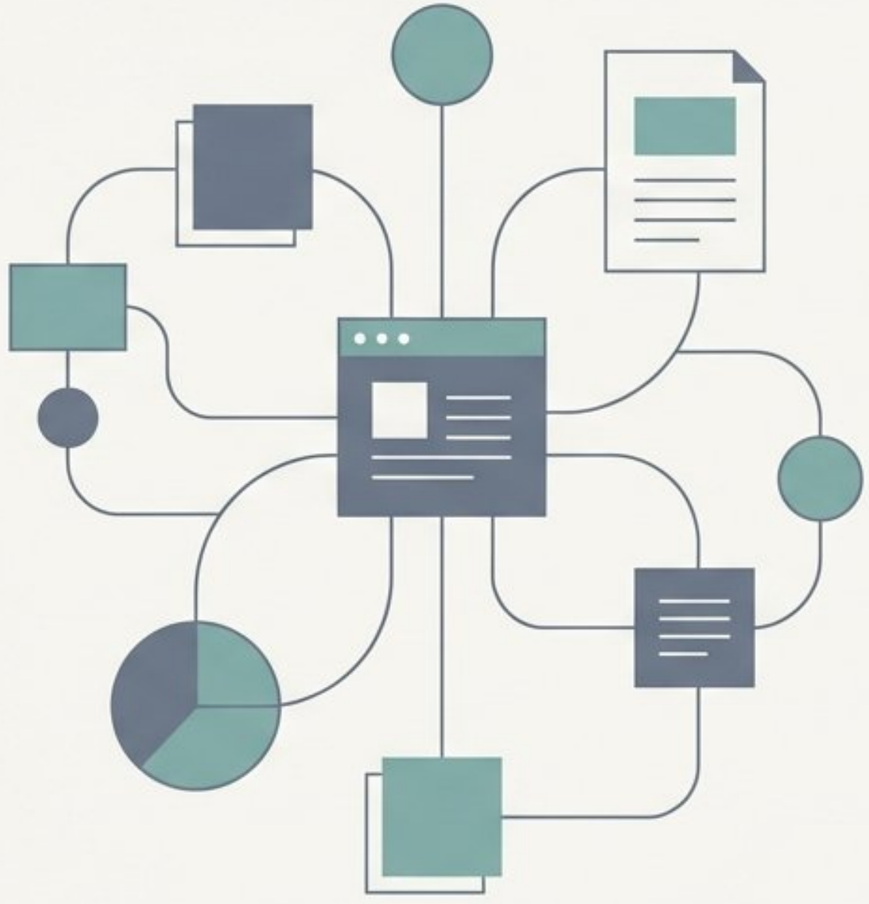
Chatbot to chat with medical books using Retrieval-Augmented Generation Model

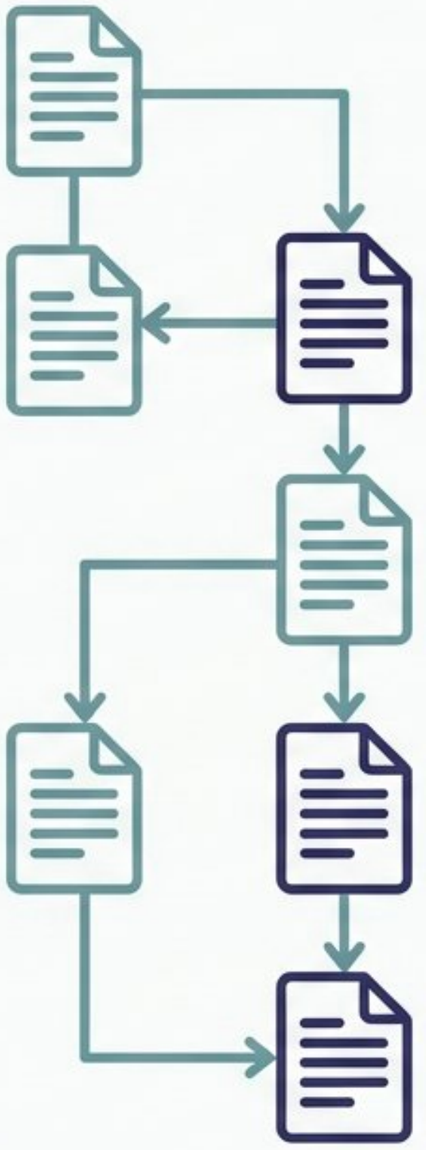


Team Members:	Guide:
• Prakhar Sharma • Akhil Kunduvalappil • Adithya Baiju • Pavithra • Devika	Amrutha.S.Aravind (Asst. Prof CSE)

NotebookLM

Presentation Content

- Abstract
- Introduction
- Literature Survey
- Problem Identification
- Proposed System
- Project Objectives
- Architecture
- Algorithm
- System and Software Requirements
- Project Plan
- Future Scope & Conclusion
- References



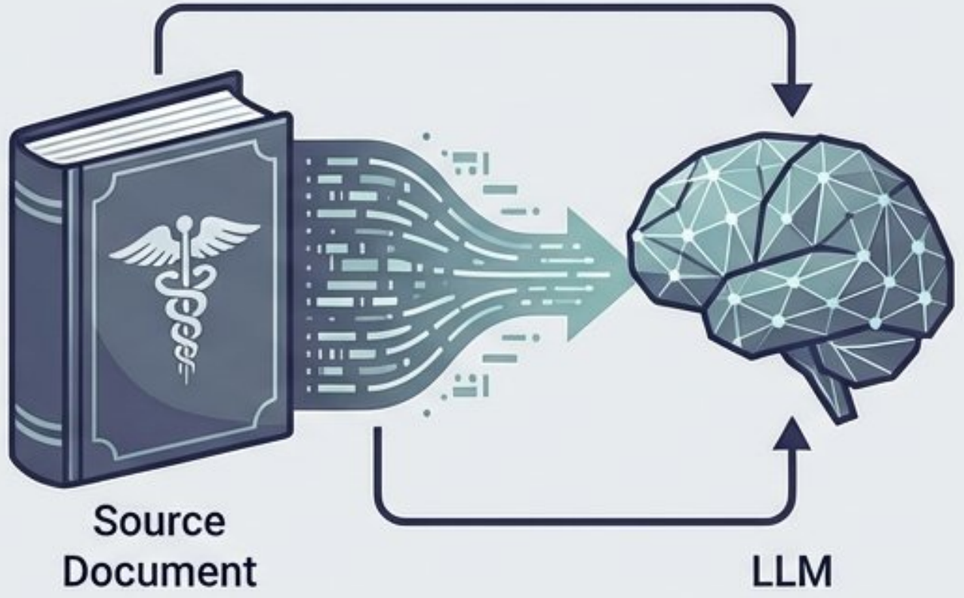
NotebookLM

Abstract

Core Technology: This project presents an AI-driven study assistant designed to relieve the massive reading workload of medical students. It uses a ***Large Language Model (LLM)**, which is the "brain" that understands and generates human language, powered by Google Gemini. To stop the AI from making up fake facts, we use ***RAG (Retrieval-Augmented Generation)**, which is a technique that gives the AI an "open book" to read from before it answers a question.

Technical Implementation: The system allows students to upload their heavy PDF textbooks. When a question is asked, it searches the book first, reads the exact paragraphs, and generates a grounded, highly accurate response.

User Experience: Through a simple Streamlit web interface, students get instant, reliable medical answers backed by their own textbooks, saving hours of searching and significantly reducing study stress.



NotebookLM

The Need for a Reliable Medical Assistant

- Medical students face an immense workload, reading thousands of pages, which leads to high stress and fatigue.
- Standard AI tools often "hallucinate" or invent fake medical facts because they lack proper context and grounding.
- MedBuddy solves this by using the Gemini API to act as a secure, local study assistant that only answers based on uploaded files.
- **PubMed**: A free search engine accessing primarily the MEDLINE database of references and abstracts on life sciences and biomedical topics. We use this concept to source legitimate medical knowledge.
- **Meditron**: A specialized dataset concept focused specifically on high-quality, verified medical and clinical guidelines.





NotebookLM

Literature Survey (Recent Works)

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
1	AI-based Mock Interview System [2024]	Ayesha Sayyad	Uses AI and NLP to assess candidate responses	General AI lacks domain-specific medical grounding and factual accuracy.
2	Role of ChatGPT and AI in Medical Education [2023]	Gunther Eysenbach	Investigated standard LLMs in medical learning	High risk of authenticity issues and "hallucinations" without a retrieval system.
3	AI Chatbots in Public Health Education [2023]	Francesco Baglivo et al.	Evaluated AI chatbots for vaccination training	Lacked professional verification and reliable source-tracking for answers.

NotebookLM

Literature Survey (Foundational Works)

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
4	Ontology Based Personal Assistive Application [2023]	N. V. Rohil et al.	Uses ontology modeling and semantic reasoning for health	Highly complex to update with new medical textbooks and literature.
5	Farm-Haystack Question Answering System [2022]	R. S. Nambiar	QA system using pre-trained BERT models	Often lacks specificity for highly specialized medical queries.
6	ChEMBL Bot for Database [2018]	V. Murali et al.	NLP chatbot to query chemical compounds	Limited strictly to a specific chemical database, cannot read full medical PDFs.

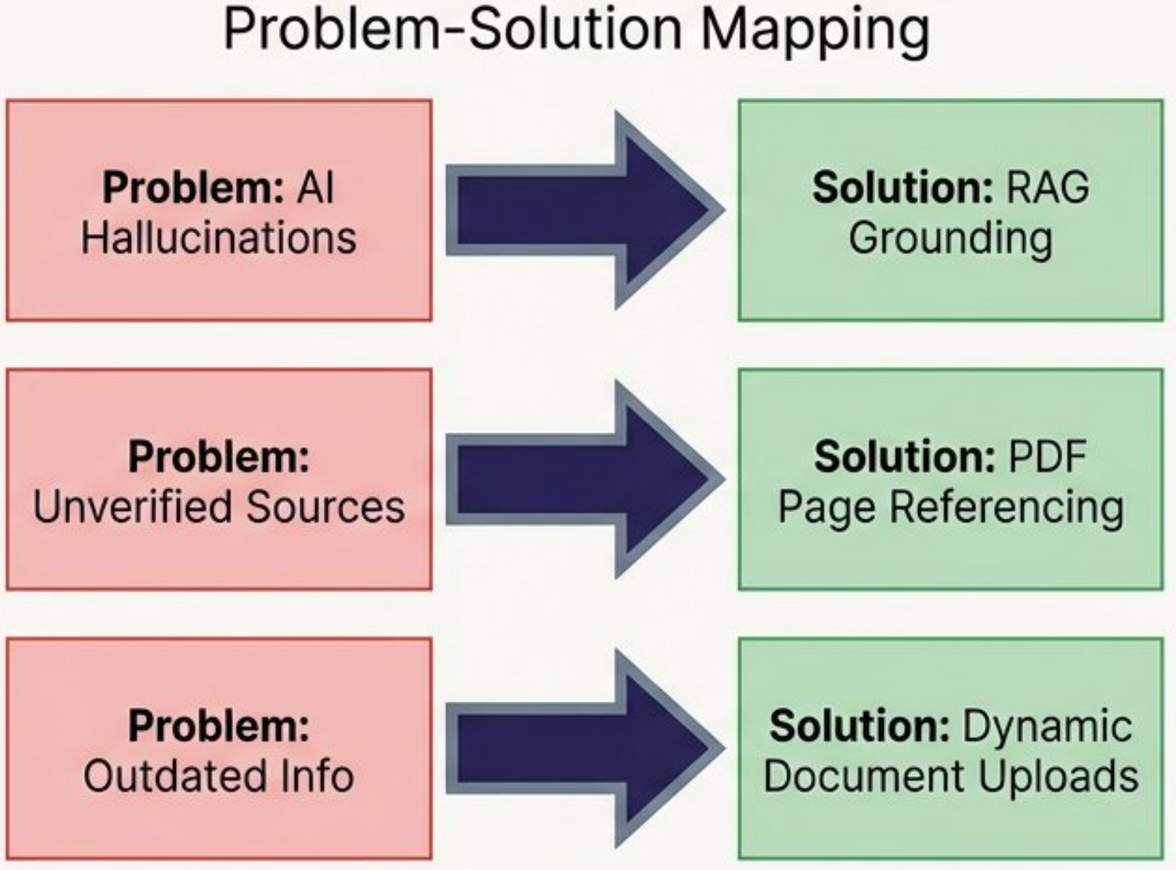
NotebookLM

Problem Identification

- **Generic AI is Unsafe:** Standard AI models guess answers when they do not know them, which is highly dangerous in medical education.
- **Lack of Source Transparency:** Existing systems do not tell the student exactly which page or book the information came from.
- **Inflexible Knowledge Bases:** Hard-coded chatbots cannot easily learn from new textbook editions without complete retraining.

Problem-Solution Mapping

- **Problem:** AI Hallucinations → **Solution:** RAG
- **Problem:** AI Hallucinations → **Solution:** RAG Grounding
- **Problem:** Unverified Sources → **Solution:** PDF Page Referencing
- **Problem:** Outdated Info → **Solution:** Dynamic Document Uploads



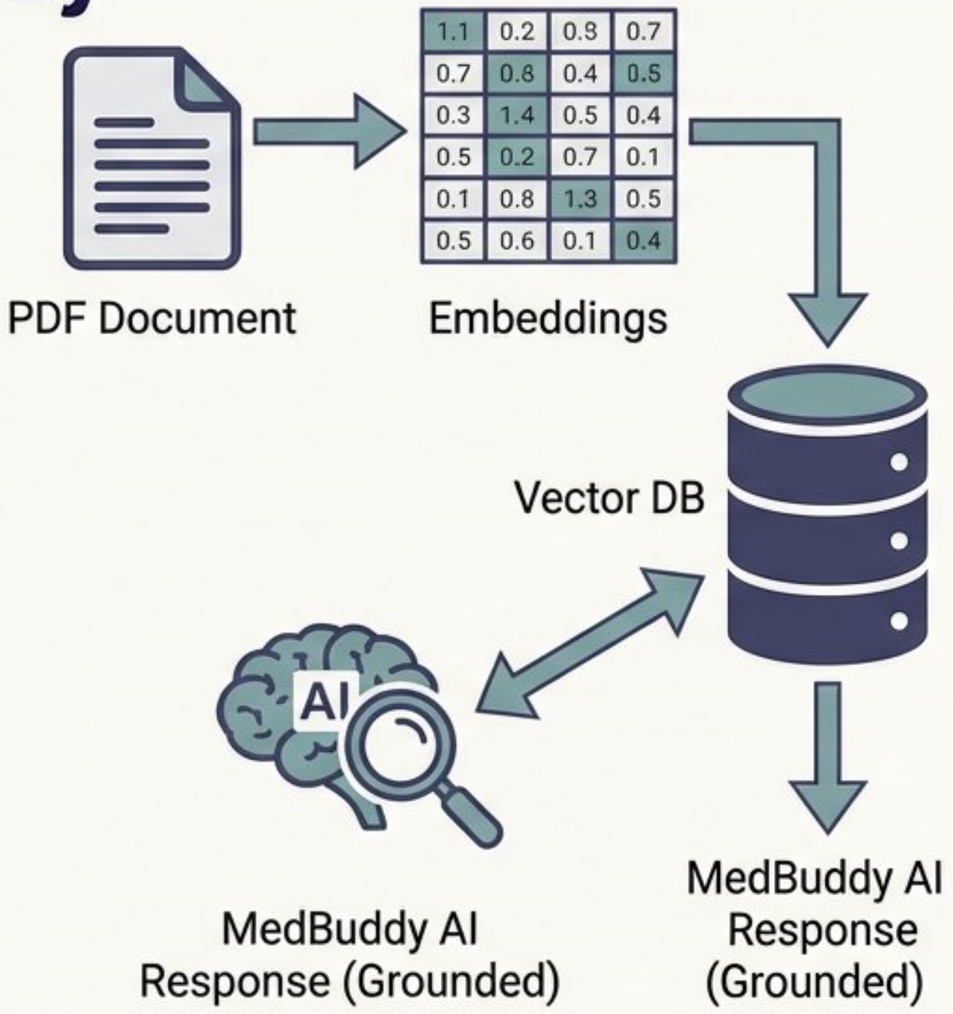
NotebookLM

The Proposed System: MedBuddy

MedBuddy uses Google Gemini as its core AI, enhanced by specific tools to ensure perfect medical accuracy derived from datasets like PubMed and Meditron.

Key Concepts:

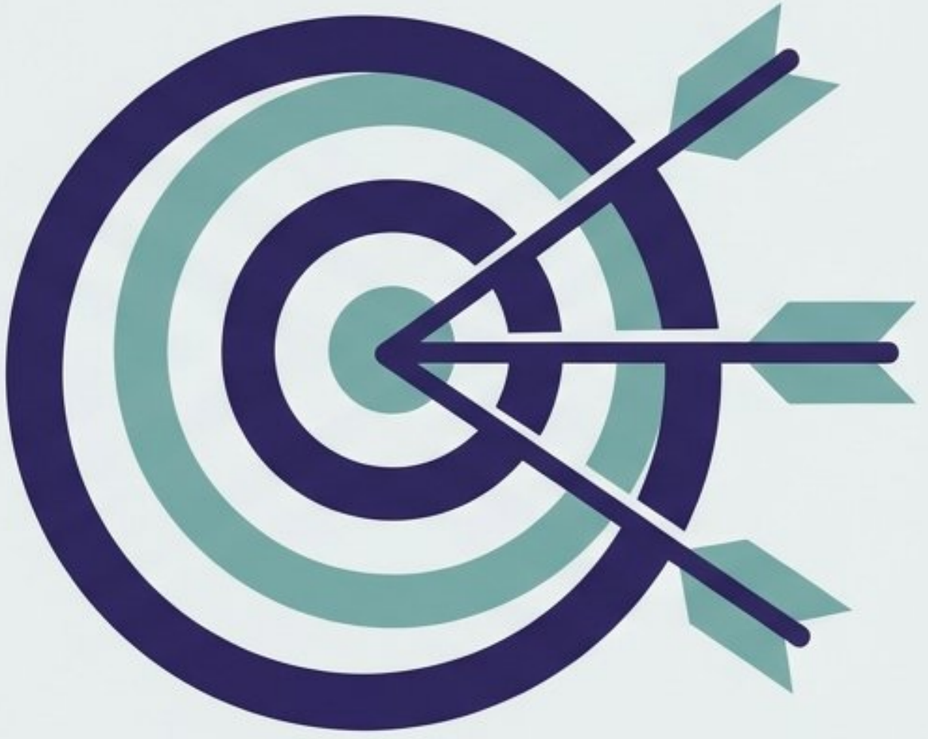
- **Embeddings**: The process of turning human words and sentences into a list of numbers so the computer can mathematically understand their meaning.
- **Vector Database**: A special storage space where we keep these "number-words." It allows the AI to instantly find paragraphs that match the user's question.
- **LLM Grounding:** A strict rule we put on the AI. We force the AI to ***only*** generate an answer using the text we provided, completely stopping it from making up outside facts.



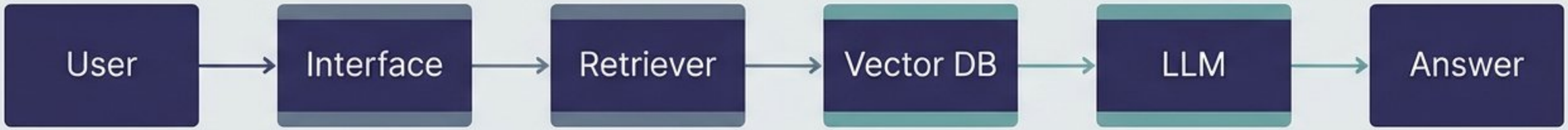
NotebookLM

Project Objectives

- **To build** an intuitive, user-friendly AI interface where medical students can upload massive textbook PDFs.
- **To retrieve** the exact, highly relevant textbook chunks instantly when a medical query is asked.
- **To generate** context-aware, highly accurate text responses based entirely on the retrieved data.
- **To improve** study efficiency and reduce the mental stress of medical students.
- **To evaluate** the system's accuracy and performance using the Trulens evaluation library.
- **To provide** grounded medical answers that are 100% free of AI hallucinations.



NotebookLM



System Architecture

The system architecture isolates the uploaded medical knowledge from the generative AI, ensuring safety and speed.

- The user interacts with the clean front-end interface.
- The system breaks the PDF into chunks and stores them mathematically.
- When asked a question, the retriever fetches only the most relevant chunks.
- The LLM reads only those chunks to formulate the final answer.

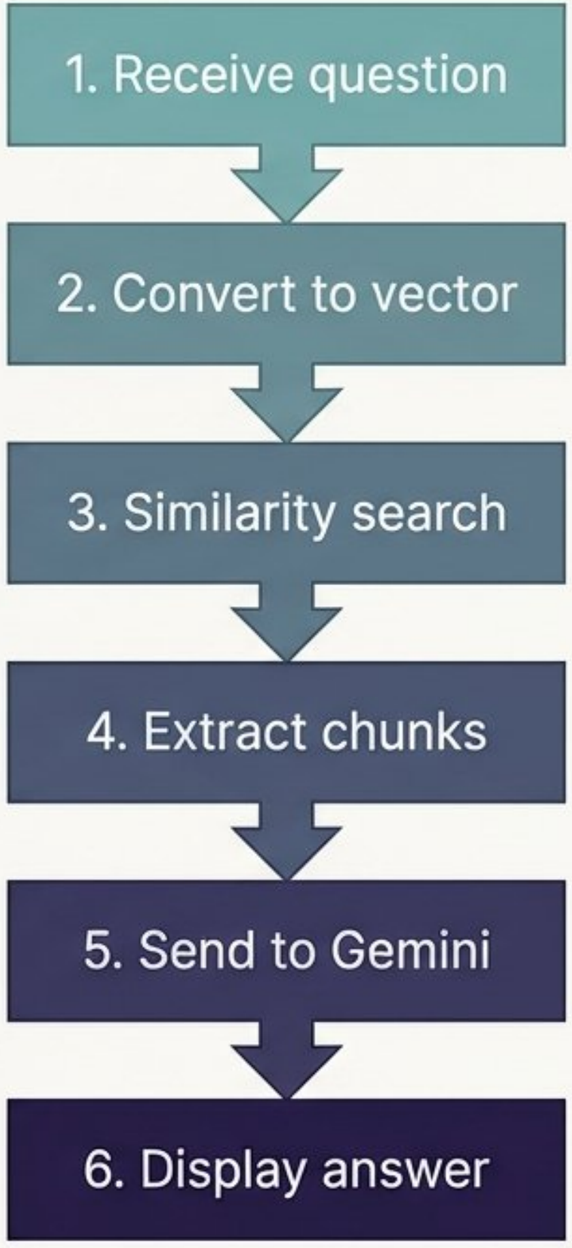
NotebookLM

Algorithm Logic

Query Processing: This is the act of taking the user's plain-English medical question, converting it into embeddings (numbers), and searching the Vector Database for paragraphs with the closest matching numbers.

Response Generation Flow: This is the final step where the retrieved textbook paragraphs and the original question are sent to the Gemini LLM. Gemini is instructed to merge this data into a readable, human-like answer.

1. Receive question from the user.
2. Convert question to vector embedding.
3. Perform similarity search in Vector DB.
4. Extract top matching textbook chunks.
5. Send chunks + question to Gemini LLM.
6. Display generated answer with source references.



NotebookLM

System and Software Requirements

Datasets & Knowledge Base

- PubMed (Medical paper
- PubMed (Medical paper formatting standard)
- Meditron (Medical data concept constraints)

Front-End

- Streamlit (Creates the web application UI)

LLM Service

- Google Gemini API

Back-End & Libraries

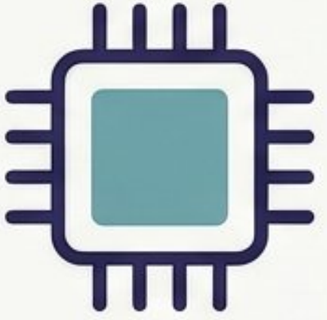
- Python (Core programming language)
- LlamaIndex (Framework to connect data to LLMs)
- Trulens (Library to evaluate AI groundedness)

Hardware Requirements

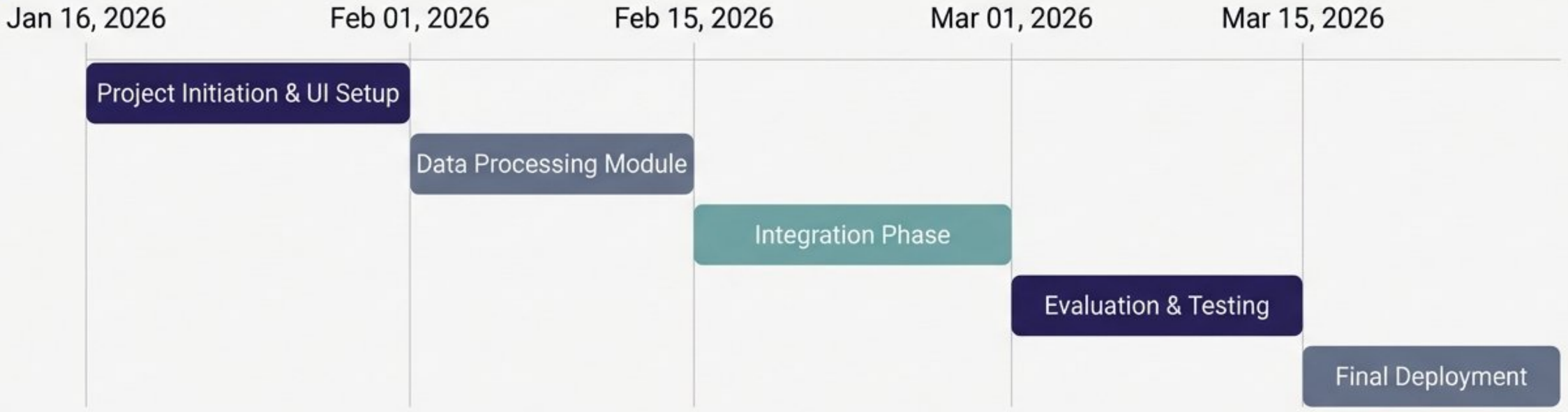
- Processor: Intel i5 or AMD Ryzen 5 minimum
- Memory: 8GB RAM minimum
- Storage: 256GB SSD minimum







NotebookLM



Project Plan

- **Jan 16, 2026:** Project Initiation & UI Setup. Building the Streamlit interface and configuring Python environments.
- **Feb 01, 2026:** Data Processing Module. Implementing PDF uploading, chunking, and embedding generation.
- **Feb 15, 2026:** Integration Phase. Connecting LlamaIndex, the Vector Database, and the Gemini API.
- **Mar 01, 2026:** Evaluation & Testing. Using Trulens to test for context relevance and groundedness.
- **Mar 15, 2026:** Final Deployment. System optimization and final presentation readiness.

NotebookLM

Future Scope

- Expanding the system to support real-time fetching from live medical journals.
- Adding voice-to-text interaction so students can study hands-free.
- Implementing user accounts to save chat histories and create personalized study flashcards.



Conclusion

MedBuddy successfully provides a safe, grounded, and hallucination-free AI assistant. By combining Google Gemini with a rigorous RAG pipeline, it achieves high accuracy (approx. 84% groundedness). It democratizes access to complex medical knowledge, significantly reducing the mental burden on students and allowing them to study smarter, not harder.

NotebookLM

References

1. Gunther Eysenbach, "The Role of ChatGPT, Generative Language Models, and Artificial Intelligence in Medical Education," JMIR Publications, 2023.
2. Francesco Baglivo, L. De Angelis, et al., "Exploring the Possible Use of AI Chatbots in Public Health Education," JMIR Publications, 2023.
3. Ayesha Sayyad, "AI-based Mock Interview System," International Journal of Research in Engineering and Science, 2024.
4. R. S. Nambiar and D. Gupta, "Dedicated Farm-Haystack Question Answering System for Pregnant Women," 12th International Conference on Cloud Computing, 2022.
5. V. Murali, R. J. Sarma, et al., "ChEMBL Bot - A Chat Bot for ChEMBL Database," 14th International Conference on Information Processing, 2018.

NotebookLM